Explicit singular-value tail bounds

This repository accompanies the paper *Sharp tail bounds for the extreme singular values of sub-Gaussian matrices*, by Marcus Webb and Morgan Lynch-Davies (in preparation).

Theorem

Let A be a random real $m \times n$ matrix, with $m \geq n \geq 1$. Write $a_i \in \mathbb{R}^n$ for its rows. Assume the rows are measurable and mutually independent. They need not have the same distribution, and coordinates within a row may be dependent.

Each row has mean zero and identity covariance: for every row i and coordinates j, k,

$$\mathbb{E}[A_{ij}] = 0, \qquad \mathbb{E}[A_{ij}A_{ik}] = \begin{cases} 1 & j = k, \\ 0 & j \neq k. \end{cases}$$

For a common parameter $K \geq 1$, assume every row satisfies the following **sub-Gaussian bound**, for every $v \in \mathbb{R}^n$ and $\theta \in \mathbb{R}$:

$$\mathbb{E}[\exp(\theta\langle a_i, v\rangle)] \leq \exp\left(\frac{K^2\theta^2\|v\|_2^2}{2}\right).$$

The coordinates, their pairwise products within each row, and the exponentials above are assumed integrable. Let $\sigma_{\min}(A)$ and $\sigma_{\max}(A)$ denote the smallest and largest singular values of A, including zero singular values, and put $L = \log(eK)$, using the natural logarithm.

For either quartet $(\alpha, \beta, \gamma, \delta)$ in the table below and every $t \geq 0$, the following bounds hold in its stated regime:

$$\mathbb{P}\Big(\sigma_{\min}(A) \leq \sqrt{m} - K\sqrt{n} - \alpha KL - \beta K\sqrt{L}\,t\Big) \leq e^{-t^2/2},$$

$$\mathbb{P}\Big(\sigma_{\max}(A) \geq \sqrt{m} + K\sqrt{n} + \gamma KL + \delta K\sqrt{L}\,t\Big) \leq e^{-t^2/2}.$$

Admissible constants

Regime	α	β	γ	δ
General: $K \geq 1$	3.84	1.58	2.35	1.50
Unit: $K = 1$	1.32	1.12	2.32	1.42

Lean statement

The complete authoritative source, MainTheoremStatement.lean, is reproduced below. These definitions state the propositions; proofs are in Canonical/Results.lean.

```
import NLAlib.Probability.SubGaussianVector.Basic
import NLAlib.Matrix.Spectral.SingularValues

open MeasureTheory ProbabilityTheory

namespace SubGaussian
universe u

def GaussianSingularValueTails {Ω : Type u} {m n : ℕ} [MeasurableSpace Ω]
    (μ : Measure Ω) [IsProbabilityMeasure μ] (A : Ω → Matrix (Fin m) (Fin n) ℝ)
    (lower upper t : ℝ) : Prop :=
    (μ {ω | NLAlib.singularValues (A ω) (n-1) ≤ lower}).toReal ≤ Real.exp(-(t^2)/2))
      ∧ (μ {ω | NLAlib.singularValues (A ω) 0 ≥ upper}).toReal ≤ Real.exp(-(t^2)/2))

def TailBoundAt (K α β γ δ : ℝ) : Prop :=
  ∀ {Ω : Type u} [MeasurableSpace Ω] (μ : Measure Ω) [IsProbabilityMeasure μ]
    (m n : ℕ) (A : Ω → Matrix (Fin m) (Fin n) ℝ),
    1 ≤ n → n ≤ m →
    (∀ i, Measurable (fun ω => A ω i)) →
    iIndepFun (fun i ω => A ω i) μ →
    (∀ i, NLAlib.IsIsotropic (fun ω => A ω i) μ) →
    (∀ i, NLAlib.IsSubgaussianVector (fun ω => A ω i) ⟨K ^ 2, sq_nonneg K⟩ μ) →
    ∀ t : ℝ, 0 ≤ t →
    let L := Real.log (Real.exp 1 * K)
    GaussianSingularValueTails μ A (√m - K * √n - α * K * L - β * K * √L * t)
      (√m + K * √n + γ * K * L + δ * K * √L * t) t

def GeneralSingularValueTailBound (α β γ δ : ℝ) : Prop :=
  (0 < α ∧ 0 < β ∧ 0 < γ ∧ 0 < δ) ∧ ∀ K : ℝ, 1 ≤ K → TailBoundAt.{u} K α β γ δ

def UnitSingularValueTailBound (α β γ δ : ℝ) : Prop :=
  (0 < α ∧ 0 < β ∧ 0 < γ ∧ 0 < δ) ∧ TailBoundAt.{u} 1 α β γ δ

end SubGaussian
```